



Course Name:	Data Communication Networks	Course Code:	EE317
Program:	Electrical Engineering	Semester:	Spring 2018
Duration:	1 Hour	Total Marks:	30
Exam Date:	12 Apr- 2018	Weight:	15
Section:	All	Page(s):	2
Exam Type:	Sessional-2		

Student Name: _____ Roll No. 14L-4445 Section: EE-3

Instruction/Notes: 1. Write your Name and Roll Numbers in above space.
2. Solve on Answer Sheets.

Question No. 1

Marks: 5+5

CLO 03 Recognize the model of MAC layer and its performance parameters.

a) We have come across three generations of Carrier Sense Multiple Access protocols – the original CSMA, CSMA/CD, and CSMA/CA. Based on your knowledge to these variants of CSMA MAC protocols, address the following questions:

i. How does CSMA work in principle?

ii. What is the problem in CSMA that CSMA/CD is trying to resolve?

iii. What is the problem in CSMA/CD which CSMA/CA is trying to resolve?

b) Suppose two nodes, A and B, are attached to opposite ends of an 800m cable, and that they each have one frame of 1,500 bits (including all headers and preambles) to send to each other. Both nodes attempt to transmit at time $t = 0$. Suppose there are four repeaters between A and B, each inserting a 20-bit delay. Assume the transmission rate is 100 Mbps, and CSMA/CD with backoff intervals of multiples of 512 bits is used. After the first collision, A draws $K = 0$ and B draws $K = 1$ in the exponential backoff protocol. Ignore the jam signal and the 96-bit time delay.

i. What is the one-way propagation delay (including repeater delays) between A and B in seconds? Assume that the signal propagation speed is 2×10^8 m/s.

ii. At what time (in seconds) is A's packet completely delivered at B?

Marks: 4+3+3

Question No. 2

CLO 04 Research the wireless networks and their applications

Give short to-the-point answers to the following questions:

i. MACA solves both hidden terminal and exposed terminal problems while CSMA/CA solves only one of these problems. Which problem CSMA/CA does not solve and Why?

ii. What is the purpose of varying inter-frame Spacings in 802.11?

iii. Why are there multiple Address fields in 802.11 frame instead of two (source MAC address and destination MAC address) fields?

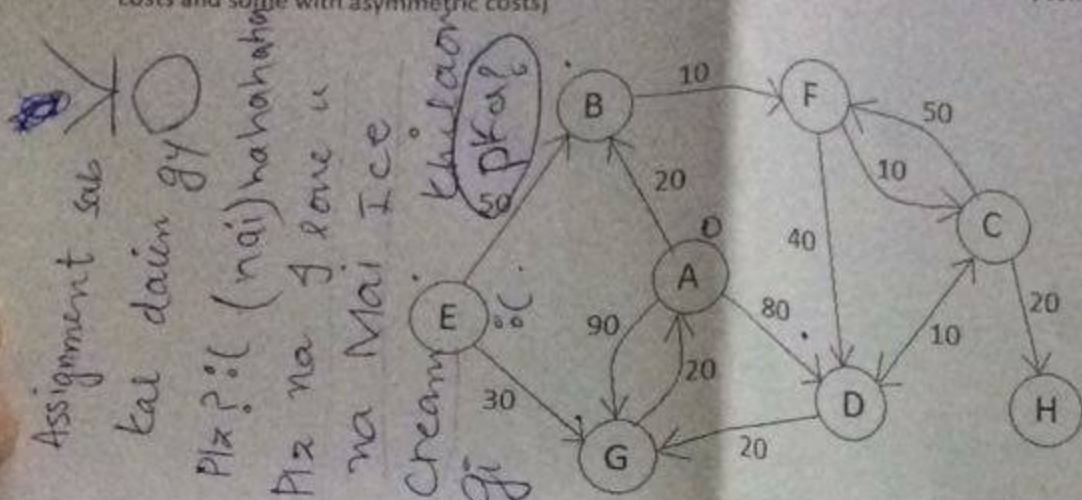
Question No. 3 (For Sections EE-1 and EE-3 only)

CLO 05 Evaluate network layer protocols and their impact on network performance.

Using Dijkstra's Algorithm, compute the shortest paths from node A to all other nodes of the network. You must write complete equations (plus put values in equations) for all iterations.

Clearly indicate which distances are updated in each iteration.

[Note: This is a directed graph with some unidirectional links, some bidirectional links; some links with symmetric costs and some with asymmetric costs]



Question No. 3 (For Section EE-2 only)

Marks: 10

CLO 05 Evaluate network layer protocols and their impact on network performance.

a) Consider the IP address 184.86.92.182 to solve the following questions.

- Suppose that we were using class-based addressing. What type of network would this IP address be a part of?
- If the network administrator decided to break this network into 8 equal subnets, how many hosts can be on this subnet? What would be the subnet mask of the subnet to which this IP address belongs?
- What is the range of IP addresses for this subnet? In other words, what is the starting IP address and ending IP address of this subnet?
- Now suppose instead that we are using CIDR addressing instead of Class-based addressing and subnets. What would be the length of CIDR prefix for the physical network in part iii) to which the host was attached?
- Suppose a packet with an IP address 184.86.31.2 arrives at the router. Does it belong to the same subnet? Justify your answer.

b) Suppose a 9000-byte IP packet is forwarded across a link with a 1500-byte MTU. How many fragments created? What are their lengths?

4000

1500

② 1480
1480

9000 - 1500 = 7500
1480
1480

CN

April 16, 2018

Mid #02

#03:

184.86.92.184

1011000.86.92.184

Class B

(i) Total?

(ii) Subnets?

Class B ka prefix 16 bits hta hai

$$32 - 16 = 16$$

$$2^{16} = 65536$$

$$65536 = 8192 \times 8$$

8 equal subnets

$$32 - 13 = 19 \rightarrow \text{prefix}$$

Subnet Mask:

$$255.255.224.0$$

(c) Range =

184.86.64.0 → first address

184.86.01011111.255 (given IP ki 19 bits k ilawa 1 kar den)

184.86.00000000.255 → last Address

Range:

$$184.86.64.0 - 184.86.95.255$$

(d) Classless Addressing k liay

Prefix length = 19

Hosts = 8192

(e) IP (given) ka 'AND' karna hai
Subnet mask k sath.

²²⁴
(184.86. ~~000~~ 0) and (184.86.31.2)

→ 184.86.0.0 (doesn't belong to the
subnet kyunke first
IP nhi same)

Q#03(b)

9000 byte = packet size

7 fragments → 6 of 1500 bytes
→ 1 of smaller size

header
 $9000 - (20) = 8980 \text{ bytes}$
data size

$\frac{8980}{1480} = 6.07 = 7 \text{ fragments}$

No. of Frag.	Total length	MF	offset
1	1500	1	0
2	1500	1	185
3	1500	1	$185 + 185 = 370$
...	"	"	"
7	$8980 - (1480 \times 6)$ $= 120$	0	100

$\frac{1480}{8} = 185$

Q1(a) i-carrier sense kartay hain

(ii) Collision Detection

CSMA/CD → After collision wali cheez

- About transmission
- Jamming signal
- Exponential back-off

(iii) - Signal strength kht kamm hojti hai in wireless signal → CSMA/CD

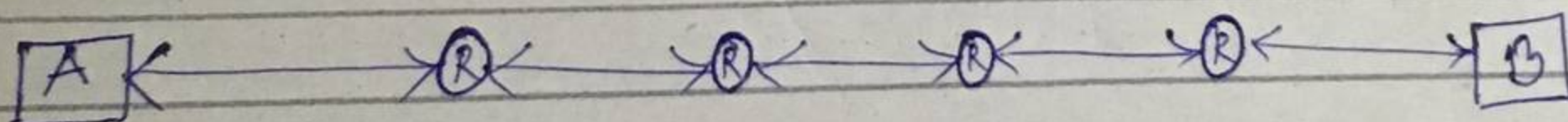
Q1(b)

cable length = 800 m

$t = 0 \rightarrow$ collision

(i)

donon ikathay transmit kar rhy hain at $t = 0$



(i) har repeater ka time delay and add

~~frame time = 1000~~

$$\text{slot time} = \boxed{\text{no. of slot}} \times \text{frame time}$$

$$0 - (2^i - 1)$$

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$d = 800 \text{ m}$$

$$v = 2 \times 10^8 \text{ m/s}$$

propagation on time (A—B):

⇒ 20 bit delay → time delay mn karna hai convert.

$$\text{propagation delay} = \left(\frac{800}{2 \times 10^8} \right) + 4 \left(\text{repeater time delay} \right)$$

$$\text{frame time} = \frac{\text{frame size}}{\text{data rate}}$$

$$= \boxed{4 \mu} + 4 \left(\frac{20}{100 \times 10^6} \right)$$

$$= 4.8 \mu\text{s} \text{ (with 4 repeaters)}$$

with no repeaters → 4 μs

512 bits delay

→ time delay nikalna hai.

$$\frac{512}{100 \text{ Mbps}}$$

collision kab hui hai? → $\frac{4.8}{2}$

detect kab hui? 1st may k band i.e. detect hui hai 4.8 μs pay and both A & B aborted Tx

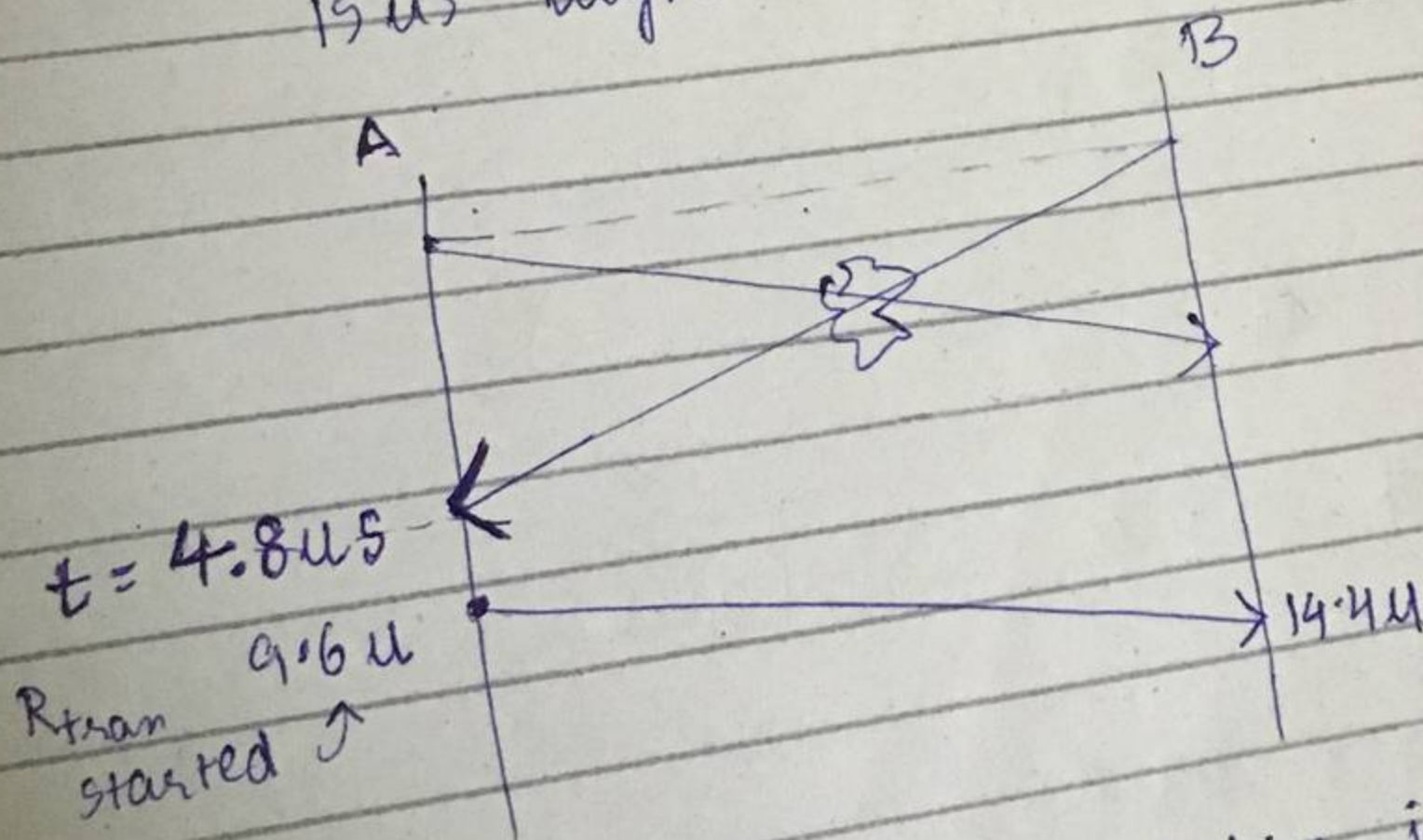
(b) cont.

$$4.8 + 4.8 = 9.6 \mu s$$

poora frame ko transmit karne me kitna time lagna hai??

$$t_T = 1500 \div 1000 \\ = 15 \mu s$$

* $4.8 \mu s$ ke baad 'B' ki bits milna shuru hogayin aur aapka frame poora transmit hne me $15 \mu s$ lagne thay



* Abort kar diya hai lekin jab tak 'A' ko sacri bits nhi milkin 'A' will not start re-transmission.
 $t = 9.6 \mu s$ pay 'B' ki last bit 'A' ko mil jayegi (aborted transmission ki)

1) A may '0' slot choose kiya tha tou it started transmission foran hi and 'B' will ~~not~~ wait. kyunke usmay slot 1 kiya hai use.

A's Retransmission ki pehli bit will reach 'B' after $4.8 \mu s$

$$4.8 + 9.6 = 14.4 \mu s$$

poora frame milnay mn $15 \mu s$ lagne thay tou

$$= 14.4 \mu + 15 \mu$$

$$= 29.4 \mu s$$

Q#02

(i) Exposed nhi hti solve

(ii) Quality of service maintain Karnay k liay

$$AIFS < DIFS$$

(iii) 4 fields hote hain because

Basic service set ~~AP~~ ^{Kay} AP k addresses k liay

BSS1
AP1

BSS2
AP2